Faculty of Computing

Year 1 Semester 2 (2025)

SE1020 – Object Oriented Programming Lab Sheet 07

Question 1

Implement a **Hotel Booking System** to manage different types of room bookings and calculate the cost of a guest's stay. The hotel offers two types of rooms: Standard Rooms and Luxury Suites. While standard rooms offer basic facilities, luxury suites include extra services like airport pickup and breakfast.

a) Create a class called **Room** with attributes for the room number, guest name, and number of nights stayed. This class should also include both a default constructor and a parameterized constructor to initialize room data.

   i. Add a method named displayRoomDetails() in the Room class that prints out all relevant room and guest details.

   ii. Implement method overloading for a method called calculateRoomCost().

   - The first version should return a default per-night cost of $150.00.
   - The second version should accept a discount per night as an integer and return the total cost after applying that discount.
     [Hint: You should call the first method inside the second to retrieve the base rate.]

b) Create a subclass called **StandardRoom** that inherits from Room and includes an additional attribute to indicate whether Wi-Fi is included.

   i. Override the calculateRoomCost() method to return the total as 100 times the number of nights.

   ii. Override displayRoomDetails() to include the Wi-Fi status and total cost.

c) Implement another subclass called **LuxurySuite** that also inherits from Room and add an attribute to indicate whether breakfast and airport pickup are included.

   i. Override calculateRoomCost() to return the total as 200 times the number of nights, plus an additional $75 if extra services are included.

   ii. Override displayRoomDetails() to reflect these details.

d) Create a class called **HotelApp** with the main method and perform followings.

   i. Create one object from the StandardRoom class using the default constructor and another object from the LuxurySuite class using the parameterized constructor.

   ii. Call both versions of the calculateRoomCost() method on these objects and display all room and guest details using the overridden methods.

Sample Output:

```
--- Standard Room ---
Room Number: 000
Guest Name: Unknown
Nights: 0
Wi-Fi Included: No
Total Cost: $0.0
Discounted Cost (per night $10 off): $0.0

--- Luxury Suite ---
Room Number: 305
Guest Name: Alice Green
Nights: 3
Breakfast & Airport Pickup: Included
Total Cost: $675.0
Discounted Cost (per night $15 off): $630.0
```

Question 02

Create an Online Food Ordering System to handle different types of meal orders and compute the total bill. The platform offers two types of meals: Regular Meals and Combo Meals. Combo meals offer additional items like drinks and desserts, while regular meals are limited to main dishes only.

a) Create a class named **Meal** with attributes for orderID (String), customerName (String), and quantity (int). Implement both a default constructor and a parameterized constructor to initialize the meal order details.

   i. Add a method displayOrderDetails() to print the basic meal order information.

   ii. Implement method overloading for a method called calculateBill():

      - One version with no parameters that returns a base price of $12.00 per meal.
      - Another version that accepts an integer couponAmount and applies a discount by subtracting the total coupon value from the final bill.

   [Hint: Use the first method to calculate the base bill before applying the discount.]

b) Create a subclass called **RegularMeal** that inherits from Meal. Add an attribute isVegetarian (boolean).

   i. Override calculateBill() to return 10.00 $\times$ quantity.

   ii. Override displayOrderDetails() to show the type of meal (Veg/Non-Veg) and total cost.

c) Create another subclass **ComboMeal** that also inherits from Meal. Add an attribute includesDrinkAndDessert (boolean).

   i. Override calculateBill() to return 15.00 × quantity + $5 if extra items are included.

   ii. Override displayOrderDetails() to show whether extra items are included and the final bill.

d) Implement a class called ResturantApp with the main method:

   i. Create one RegularMeal object using the default constructor.

   ii. Create one ComboMeal object using the parameterized constructor.

   iii. Call both versions of calculateBill() for the objects.

   iv. Display all order details using displayOrderDetails().

Sample Output:

```
--- Regular Meal ---
Order ID: 000
Customer Name: Guest
Quantity: 0
Meal Type: Vegetarian
Total Bill: $0.0
Bill with $2 coupon: $0.0

--- Combo Meal ---
Order ID: C102
Customer Name: Alice Green
Quantity: 2
Includes Drink & Dessert: Yes
Total Bill: $35.0
Bill with $3 coupon: $29.0
```